

ENGINEERING SPECIFICATIONS

Document Name: Konnect Installation Manual		
Document No: ESP-19-20-008	Revision: 01	Date:09/09/2021
Prepared By: CGH	Approved By: RDT	Sheet: Page 1 of 23

QFlash User Guide

Rev. QFlash_User_Guide_V3.4

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Firmware Upgrade Procedures

The firmware can be upgraded through the following three steps by the QFlash tool.

Step 1: Set serial port and baud rate.

Step 2: Load firmware files.

Step 3: Upgrade the firmware.

The following describes the details of how to use the tool to upgrade firmware.

1.1. Configure Serial Port and Baud Rate

When the QFlash tool is opened, the main interface will show as below.

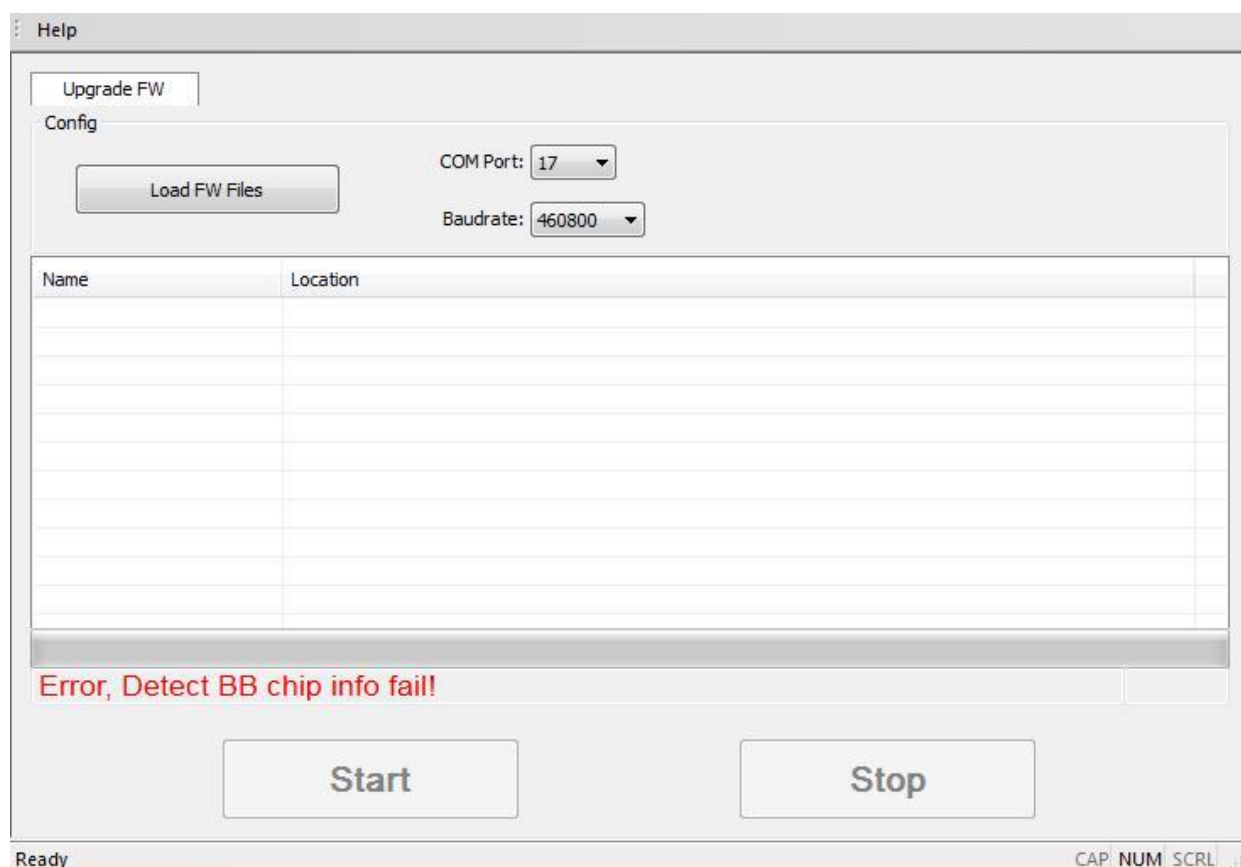


Figure 1: Main Interface

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1.1.1. Set Serial Port

1.1.1.1. COM Port Selection for M66 Modules

Click “**COM Port**” dropdown list to select the COM port through which the firmware is upgraded. As shown in the following figure.

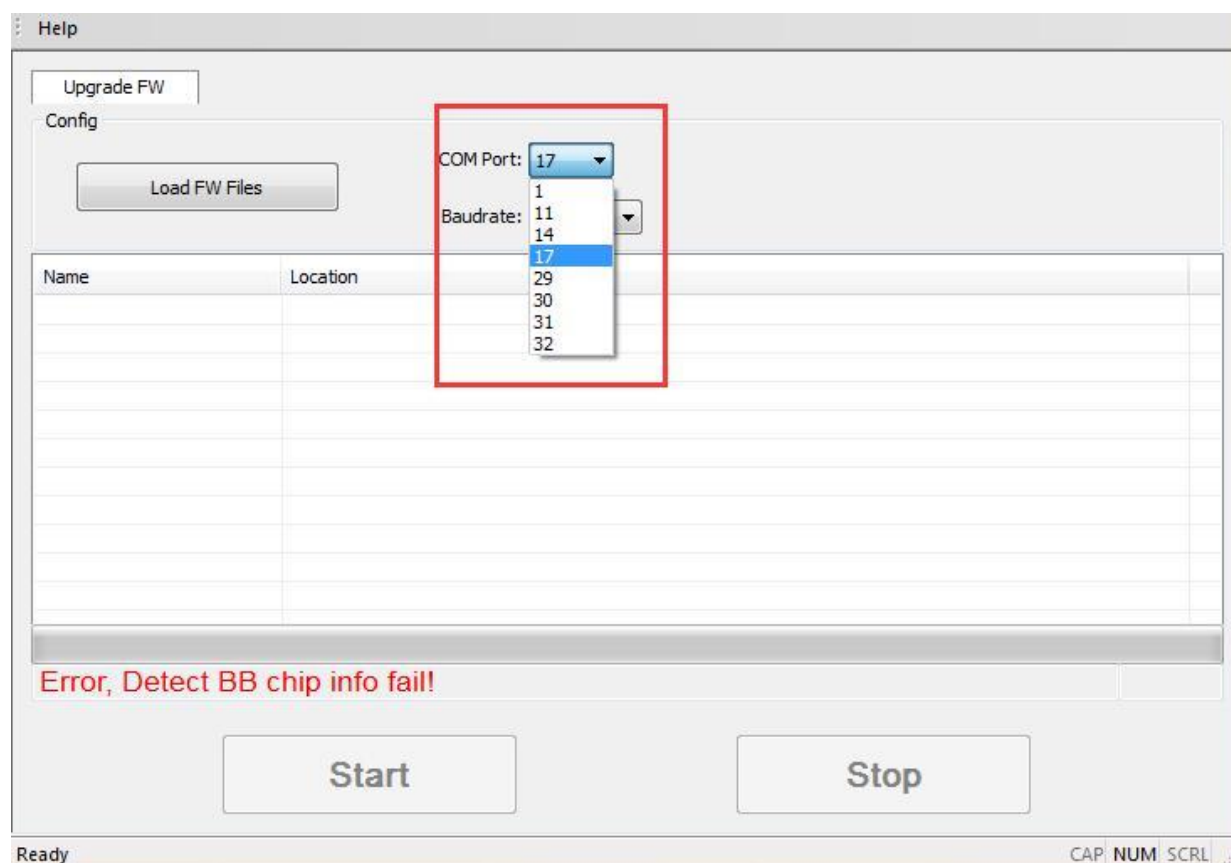


Figure 2: Select the Correct Serial Port for Mxx/GCxx/BCxx Modules

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1.1.2. Set Baud Rate

Click the “**Baud rate**” dropdown list and select an appropriate baud rate. It is recommended to select 460800 for M66 other Quested modules, as shown in the following figure.

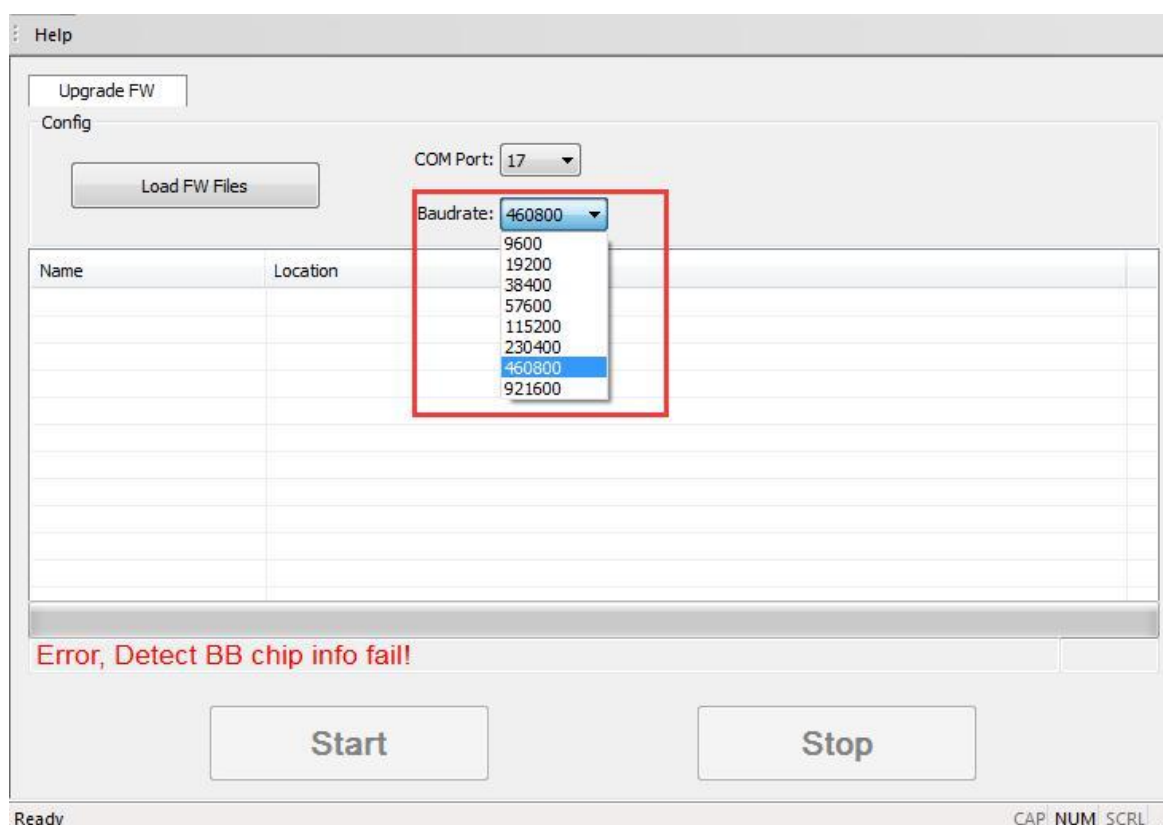


Figure 3: Select the Baud Rate

NOTES

1. Baud rates have many different values, and it is the hardware environment that determines whether a specified baud rate can be supported. If not supported, an error message will be returned.

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1.2. Load Firmware Files

1.2.1. Load APP Firmware for Open CPU or Quecopen Modules

This step is only necessary for Quectel Open CPU or Quecopen modules.

Step 1: Click the button “Load FW Files” and select the **.cfg** file which needs to be downloaded to the module.

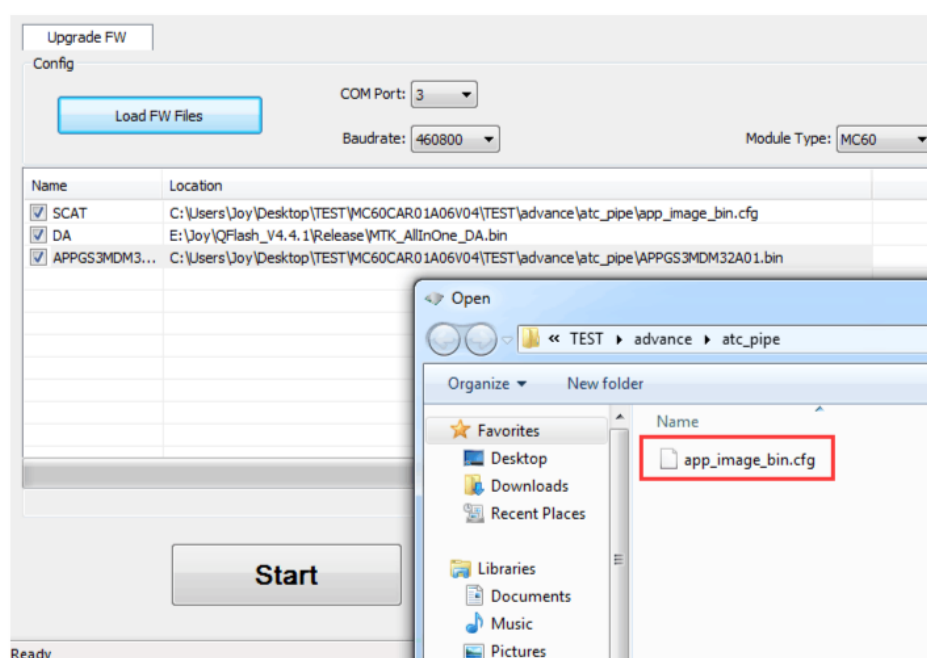


Figure 4: Select the .cfg File

NOTE

The path where the firmware is stored should NOT contain any spaces, and English characters are preferred.

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Step 2: Click the “**Module Type**” dropdown list and choose an appropriate module type. (M66)

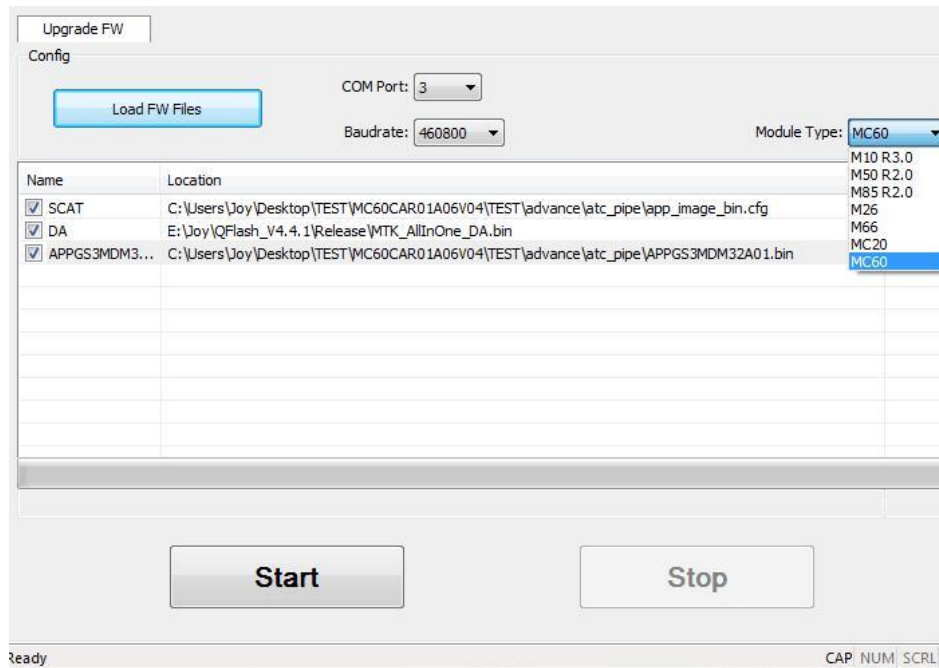


Figure 5: Select the Module Type

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1.3. Upgrade the Firmware

Step 1: before Click to “**Start**” button to upgrade the firmware disconnect module power supply. after clicking “**Start**” button then connect the module power supply. It will start to upgrade the firmware as shown in the following figure.

Upgrade FW

Config

Load FW Files

COM Port: 29

Baudrate: 460800

Name	Location
☒ 0:aboot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\appsboot.mbn
☒ 0:boot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:modem	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\NON-HLOS.ubi
☒ 0:recovery	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:recoveryfs	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:sys_back	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:system	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:SBL	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\sbl1.mbn
☒ 0:recoveryfs_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:qdsp6sw_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\NON-HLOS.ubi
☒ 0:usr_data	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\usrdata.ubi

Do not remove USB or terminate the downloading process before it completes.

Start

Ready

CAP NUM SCRL

Figure 6: Click the Start Button

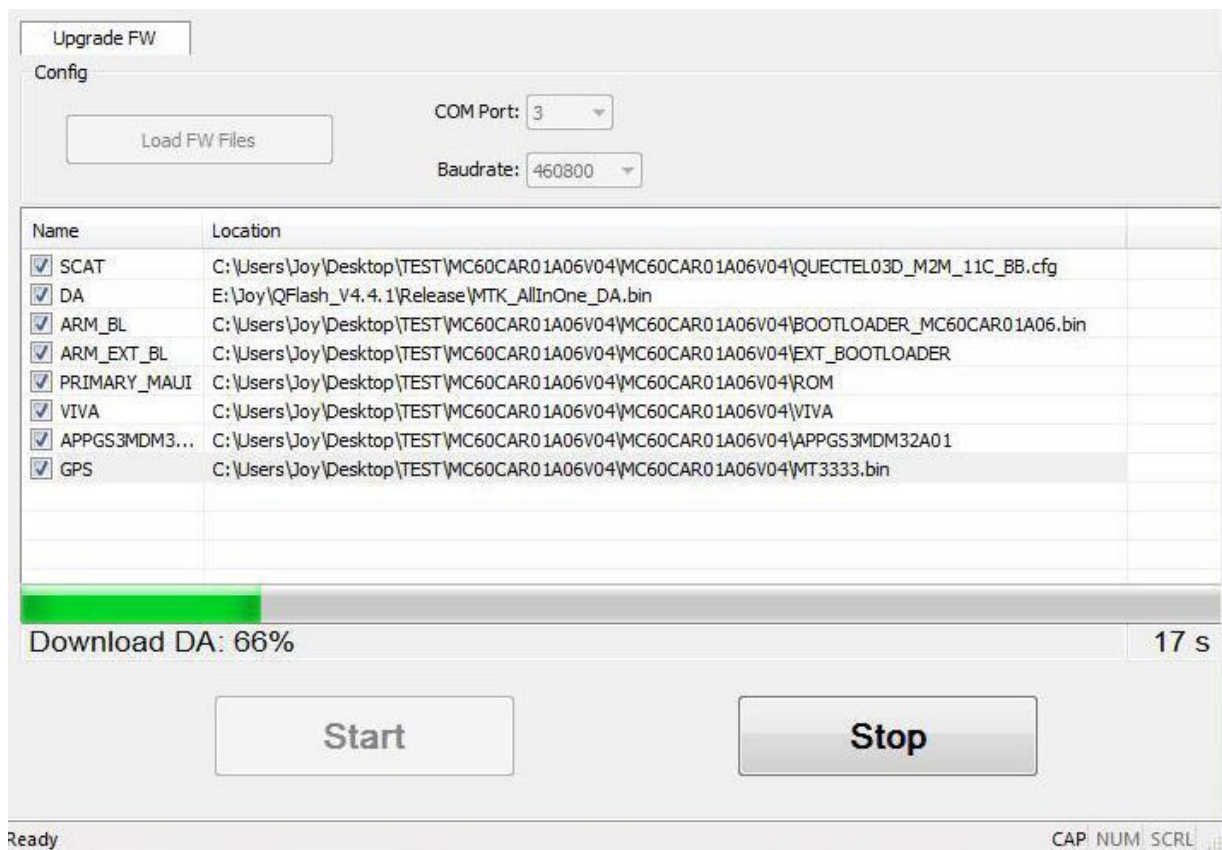
NOTES

- Please note that it is NOT permitted to stop the upgrading process, and please do NOT remove USB or terminate the downloading process before upgrading is completed.

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Step 2: It will start to upgrade the firmware as shown in the following figure.



Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAUI	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

Download DA: 66% 17 s

Start Stop

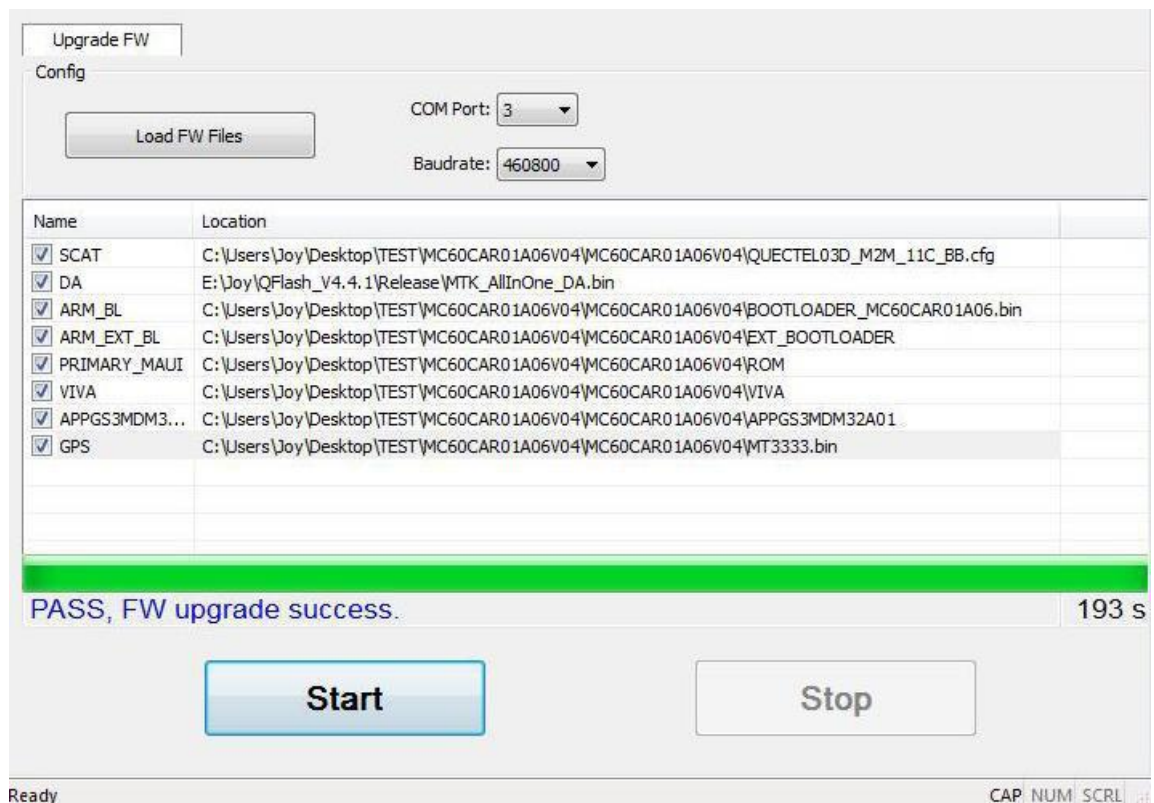
Ready CAP. NUM. SCRL

Figure 7: Start to Upgrade after Restarting the Mxx Modules

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Step 3: “FW upgrade success” will be shown on the interface after the firmware has been successfully upgraded, as shown in the following figure.



The screenshot shows a software interface for upgrading firmware. At the top, there is a tab labeled 'Upgrade FW'. Below it, a 'Config' section contains a 'Load FW Files' button, a 'COM Port' dropdown set to '3', and a 'Baudrate' dropdown set to '460800'. A table lists various firmware components with checkboxes and their file locations. All components are checked. Below the table, a green progress bar is visible. At the bottom, a status bar displays 'PASS, FW upgrade success.' followed by a timer showing '193 s'. Two large buttons, 'Start' and 'Stop', are positioned below the status bar. The bottom of the window shows a 'Ready' status and a small panel with 'CAP', 'NUM', and 'SCRL' indicators.

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAU1	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

PASS, FW upgrade success. 193 s

Start Stop

Ready CAP NUM SCRL

Figure 8: Successful Upgrade

After selecting a correct serial port, if the Mxx modules are not restarted, then the error message will be the same as that of selecting a wrong serial port.

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1.4.2. Connected to an Occupied Serial Port

Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAU1	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

FAIL, COM:3, FlashToolError:1011, S_COM_PORT_OPEN_FAIL

Start Stop

Ready CAP NUM SCRL

Figure 10: Connected to an Occupied Serial Port (Mxx Modules)

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1.4.3. Selected an Unsupported Baud Rate

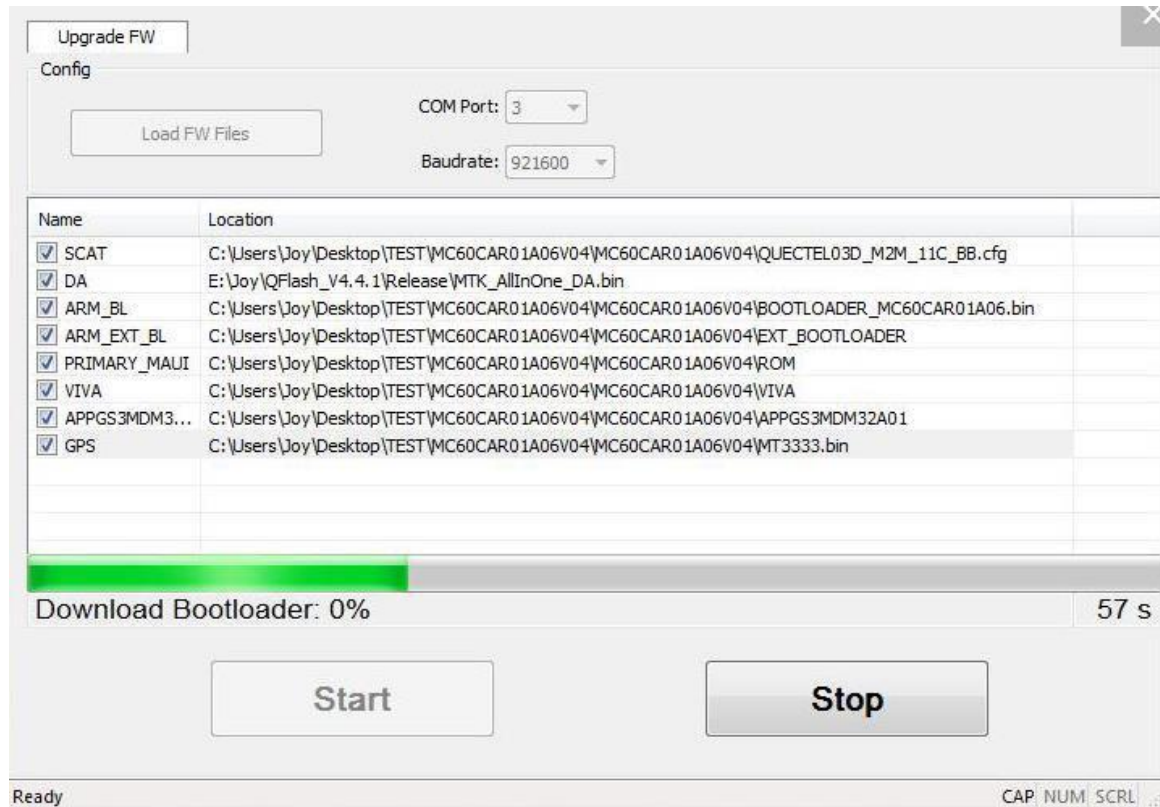


Figure 11: An Unsupported Baud Rate is Selected (Mxx Modules)

NOTE

For Mxx modules, if an unsupported baud rate is selected, the tool will stop to run and no error message will be prompted. In such case, please click the “**Stop**” button to re-select a supported baud rate to upgrade.

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1.4.4. Selected an Invalid Load File

The screenshot shows the 'Upgrade FW' application window. It has a 'Config' section with a 'Load FW Files' button, a 'COM Port' dropdown set to '3', and a 'Baudrate' dropdown set to '460800'. Below this is a table with two columns: 'Name' and 'Location'. The table contains two rows: 'RamRun' with location 'E:\Joy\QFlash_V4.4.1\Release\ramrun\host_8809_fish_spi32m_ramrun.lod' and 'FW' with location 'C:\Users\Joy\Desktop\TEST\GC10RCR02A10V01\GC10RCR02A10.lod'. Both rows are checked. At the bottom, a red error message reads 'FAIL, Read rsp pkt, timeout, no data!' with a '9 s' timer. A 'Start' button is also visible.

Name	Location
☒ RamRun	E:\Joy\QFlash_V4.4.1\Release\ramrun\host_8809_fish_spi32m_ramrun.lod
☒ FW	C:\Users\Joy\Desktop\TEST\GC10RCR02A10V01\GC10RCR02A10.lod

FAIL, Read rsp pkt, timeout, no data! 9 s

Start

Ready CAP. NUM. SCRL

Figure 12: An Invalid Scatter File is Selected (Mxx Modules)

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1.4.5. Power Supply is Abnormal

Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_v4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAU1	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

FAIL, COM:3, FlashToolError:4006, S_FT_CHANGE_BAUDRATE_FAIL

25 s

Start

Stop

Ready

CAP NUM SCRL

Figure 13: Power Supply is Abnormal (Mxx Modules)

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1.4.6. USB to RS-232 Converter Cable is Abnormal

The screenshot shows the 'Upgrade FW' software interface. At the top, there is a 'Config' section with a 'Load FW Files' button, a 'COM Port' dropdown set to '29', and a 'Baudrate' dropdown set to '460800'. Below this is a table listing files to be upgraded:

Name	Location
☒ 0:aboot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\appsboot.mbn
☒ 0:boot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:modem	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\WON-HLOS.ubi
☒ 0:recovery	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:recoveryfs	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:sys_back	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:system	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:SBL	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\sbl1.mbn
☒ 0:recoveryfs_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:qdsp6sw_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\WON-HLOS.ubi
☒ 0:usr_data	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\usrdata.ubi

Below the table, a red error message is displayed: 'FAIL, Send_Mbn_Pkt_Data_V3 timeout [Send_Mbn_Pkt_Data_V3] 37 s'. At the bottom, there is a large 'Start' button. The status bar at the very bottom shows 'Ready' on the left and 'CAP NUM SCRL' on the right.

Figure 14: USB Converter Cable is Abnormal

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Docklight User Guide

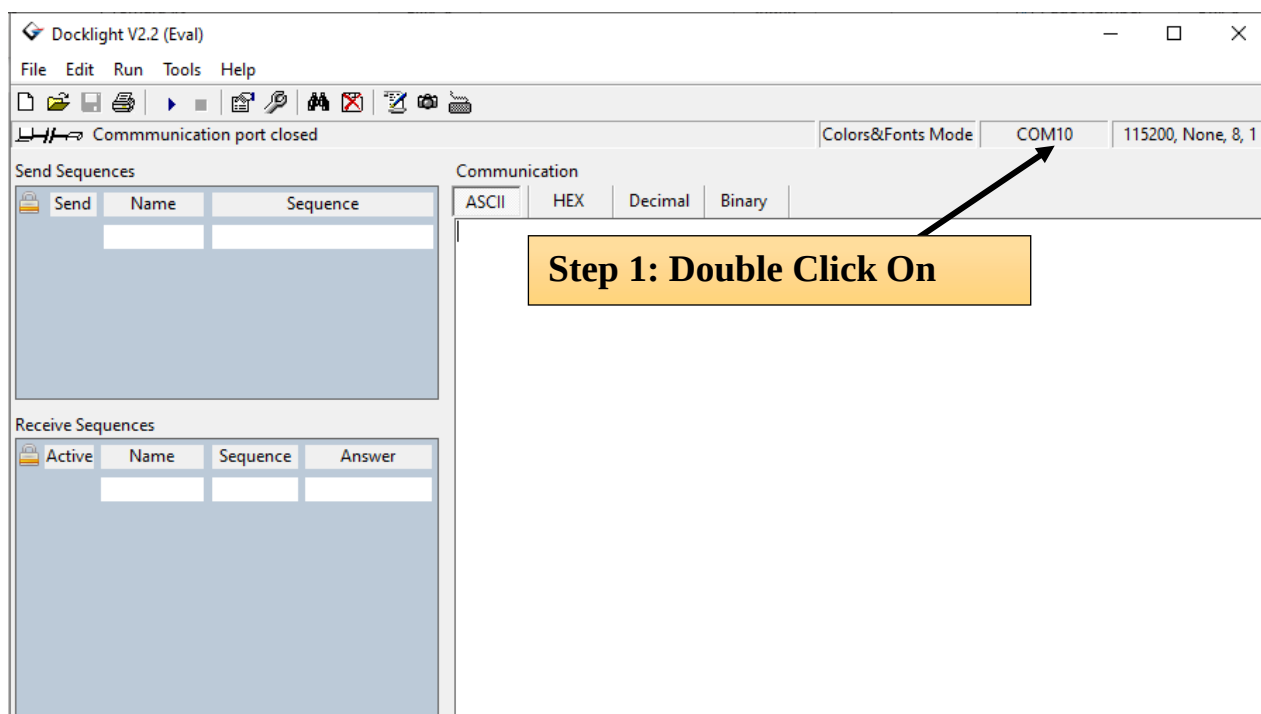
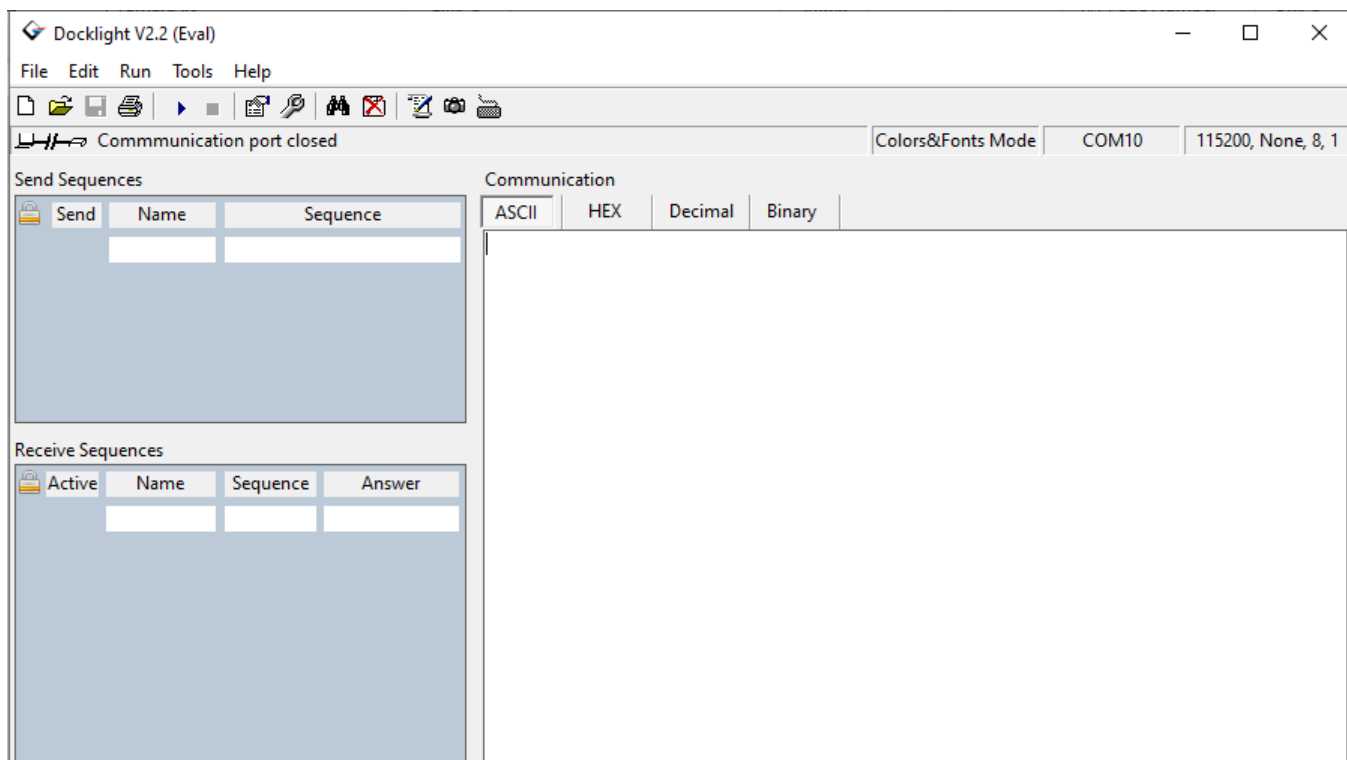
Rev. Docklight_V2.2.8

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Configure Serial Port and Baud Rate

When the docklight tool is opened, the main interface will show as below.



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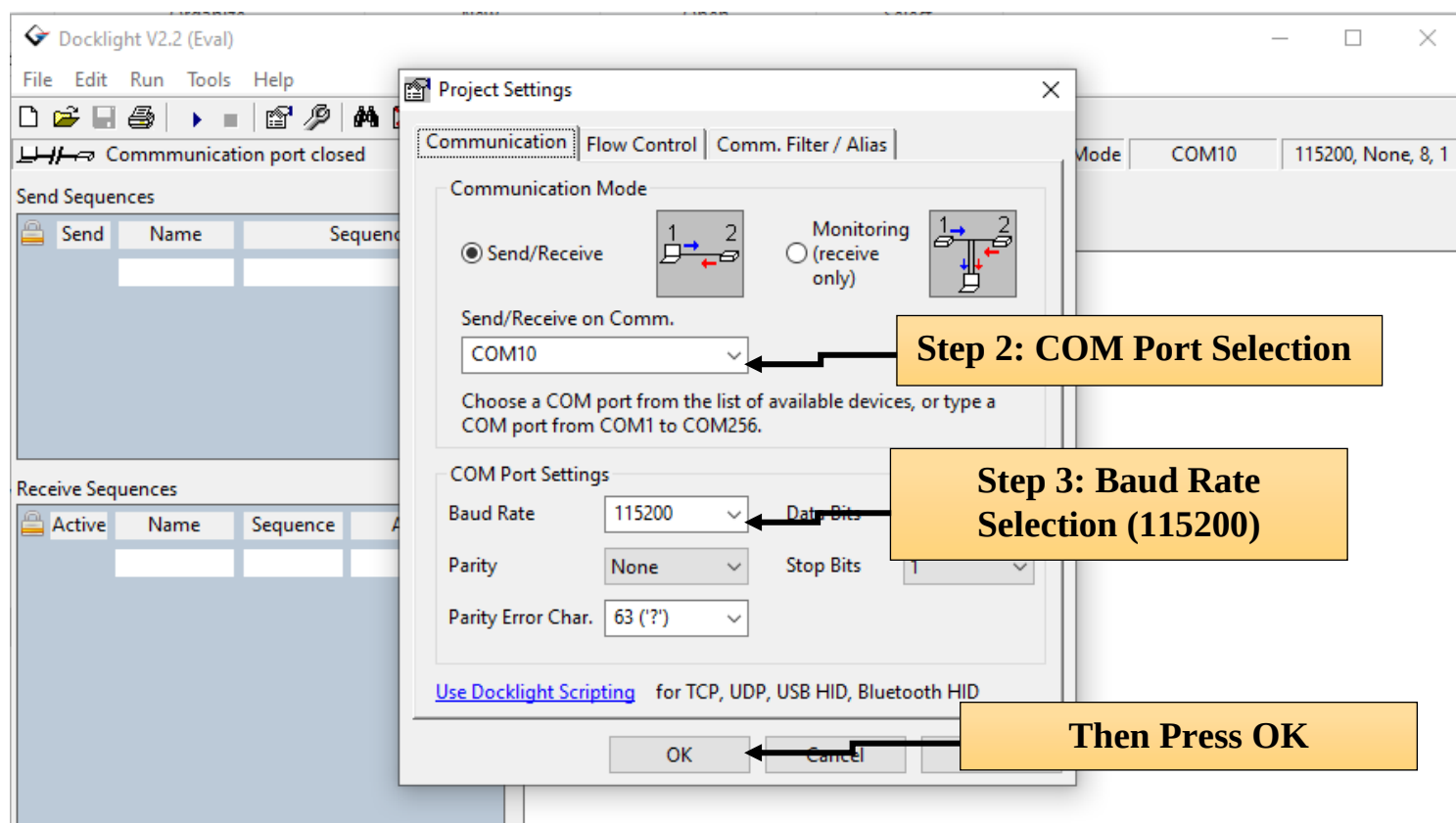
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Then,

Step 2: Select Com Port which one you are using.

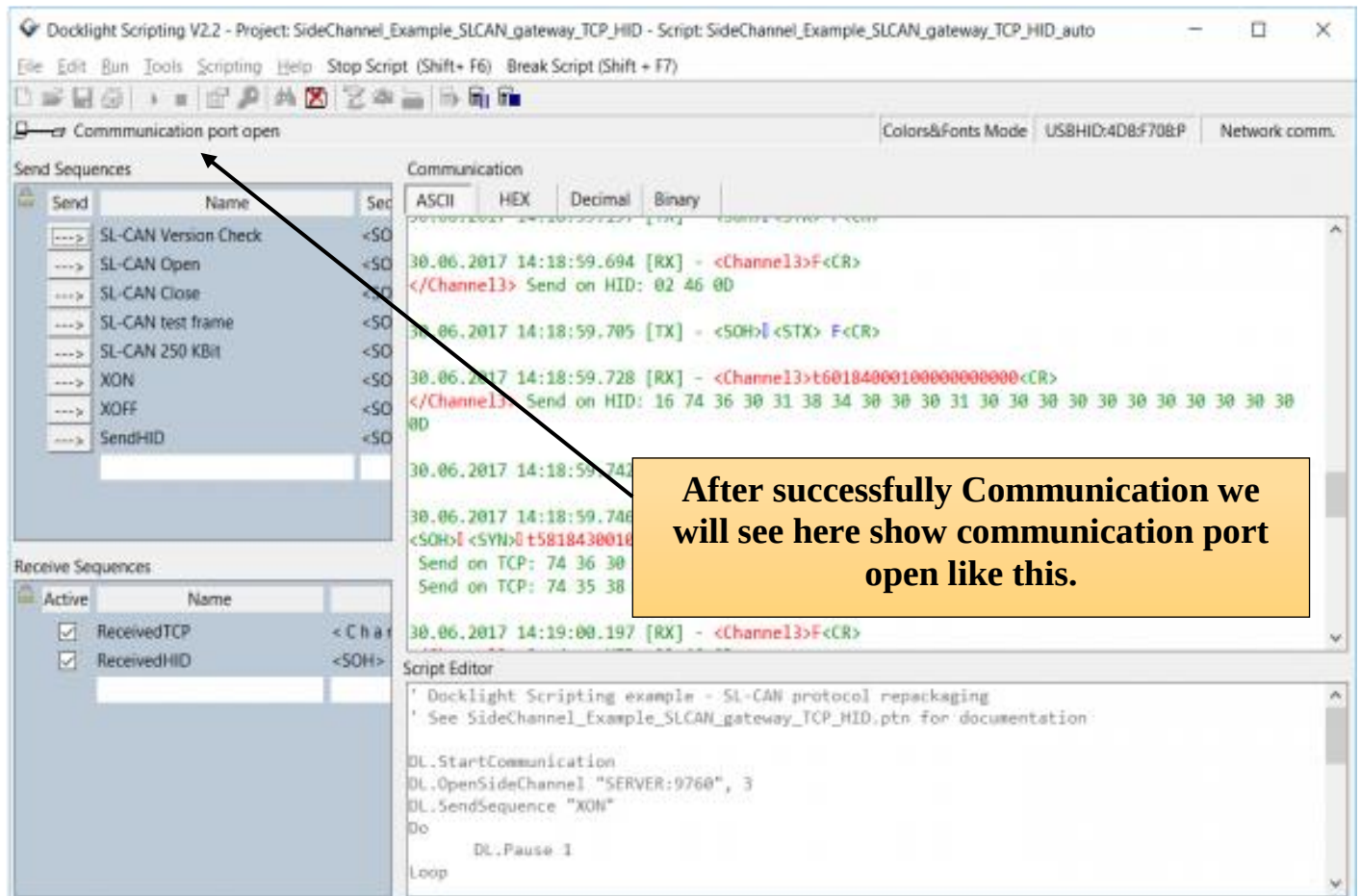
Step 3: Select Baud Rate 115200 For PC to Konnect Device.

Then press OK.



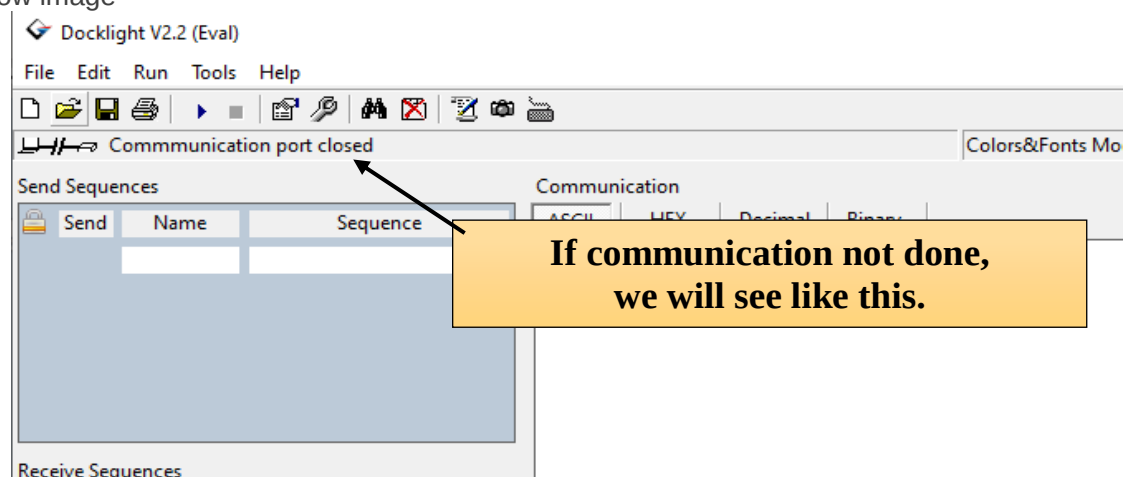
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If there is **Communication Port closed** then communication between PC and Konnect device not done,
Need to check USB cables or COM port or baud rate.

Ex. See below image



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After Successfully communication we need to set device ID to Connect Device as per our Konnect identification plan.

As shown Below table in Konnect protocol folders you will see below 5 folders P1, P2, P3, P4, P5, P6, P7, P8, P9.

See below Fig.

K-Connect Installation Plan										
Oracle Part Number		Device Identification Region Wise								
GSM Based	Ethernet Based	Delhi	Kolkata	Gujrat	Mumbai	Pune	Bangluru	Chennai	Hyderabad	Overseas
56.137.50.411	56.137.50.412	XX-1-XXXXX	XX-2-XXXXX	XX-3-XXXXX	XX-4-XXXXX	XX-5-XXXXX	XX-6-XXXXX	XX-7-XXXXX	XX-8-XXXXX	XX-9-XXXXX

Device Identification Discription										
Protocol	Devic ID	Chiller Model	X	X	X	X	X	X	X	X
			Devic ID Chiller Model Wise		Region Identificati	Device Number Setteble While Programing				
Folder P1	80	KWK/KWI Single Circuit	8	0						
Folder P2	81	KWK /KWI Double Circuit	8	1						
Folder P3	82	KWS/KAS Single Circuit	8	2						
Folder P4	83	KWS/KAS Double Circuit	8	3						
Folder P5	84	KSC Unit	8	4						
Folder P6	85	KCM Single Circuit	8	5						
Folder P7	86	KCM Double Circuit	8	6						
Folder P8	87	KCM Triple Circuit	8	7						
Folder P9	88	KSM Unit	8	8						

For EX... If you are installing first device For KSC Single Ckt (Pune Branch).

Your address will be like this.

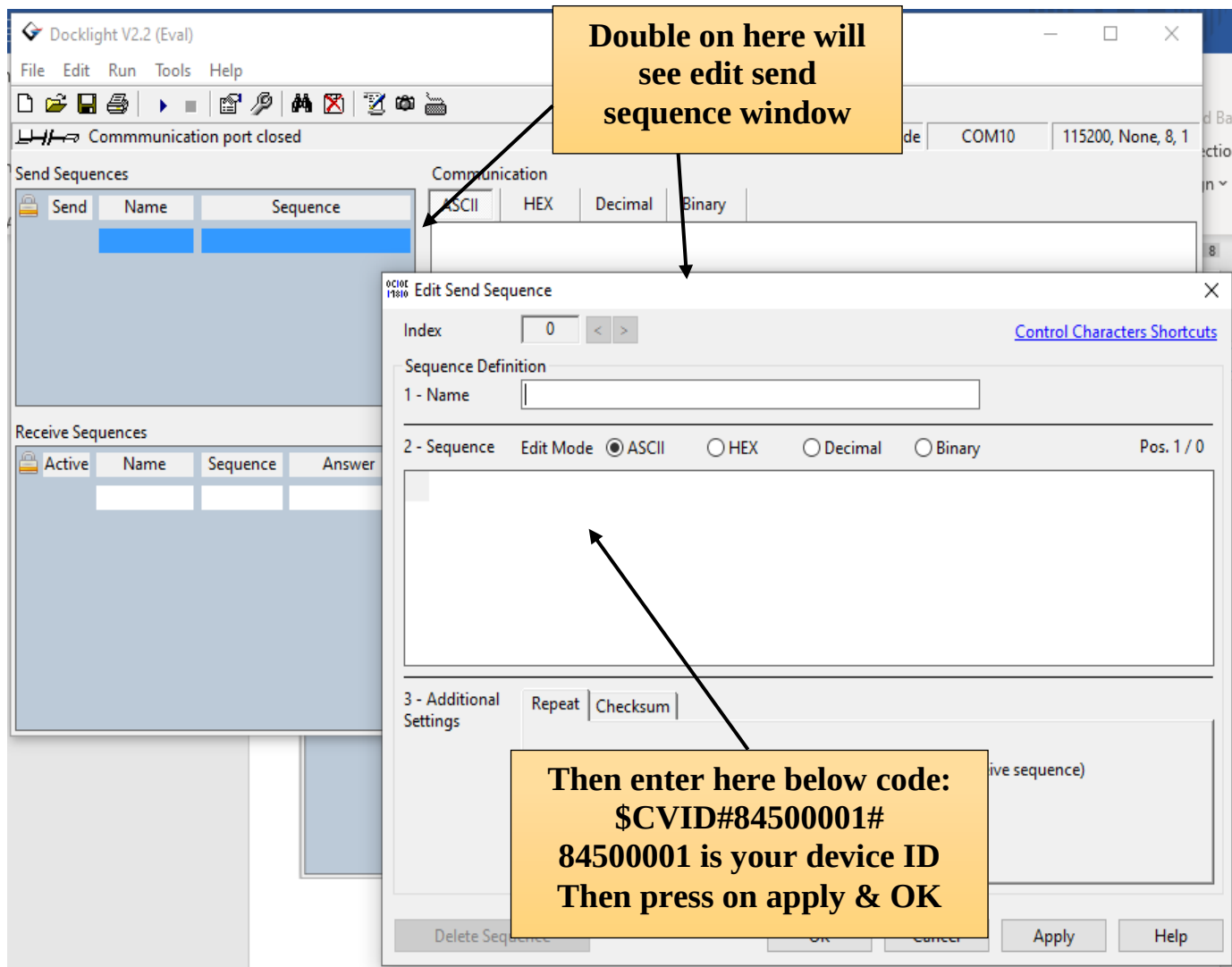
Device ID Chiller Model Wise		Region Identification	Device Number settable while programing				
8	4	5	0	0	0	0	1

Note- For overseas Konnect installation use 9 as region identification.

Now after creation of device ID need to assign device id to Konnect device, as show below image,

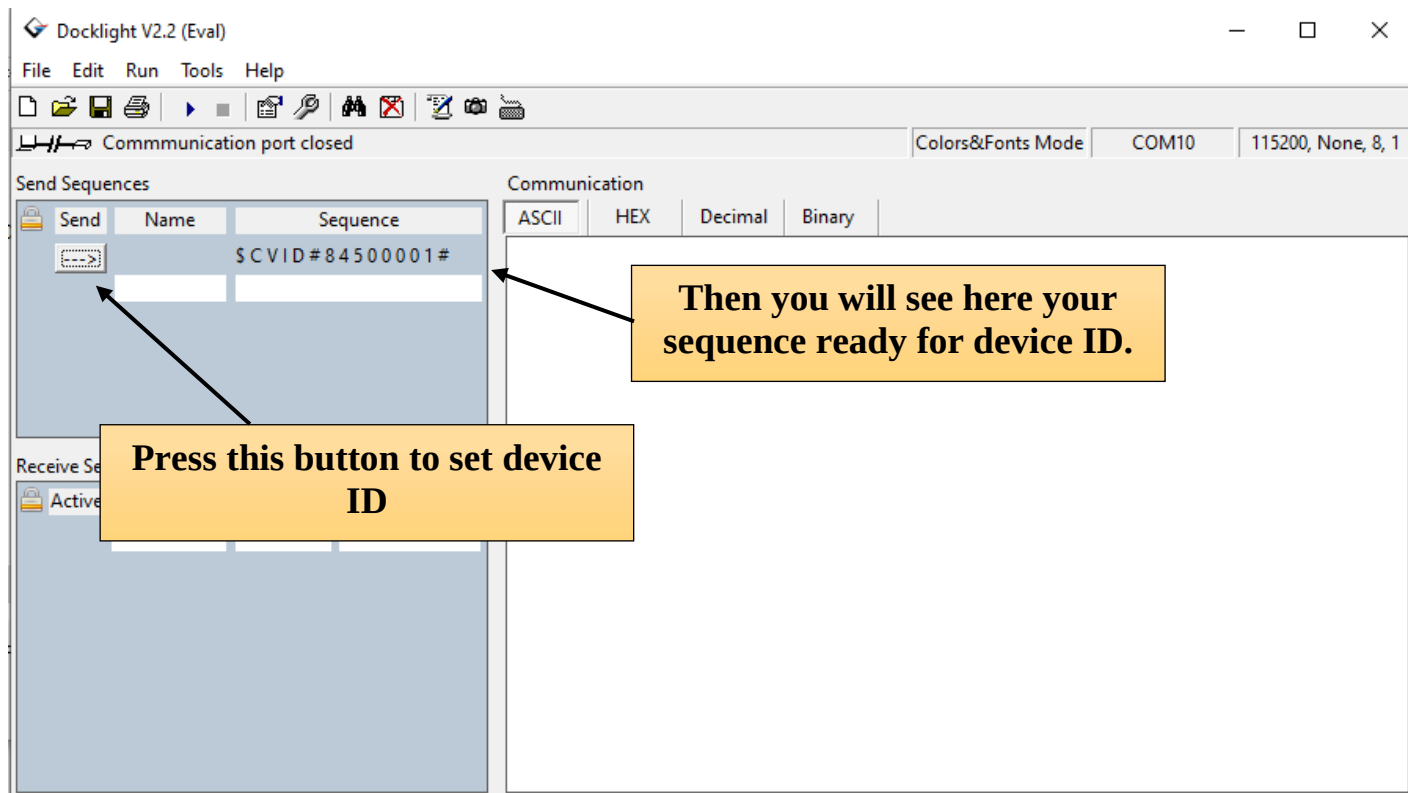
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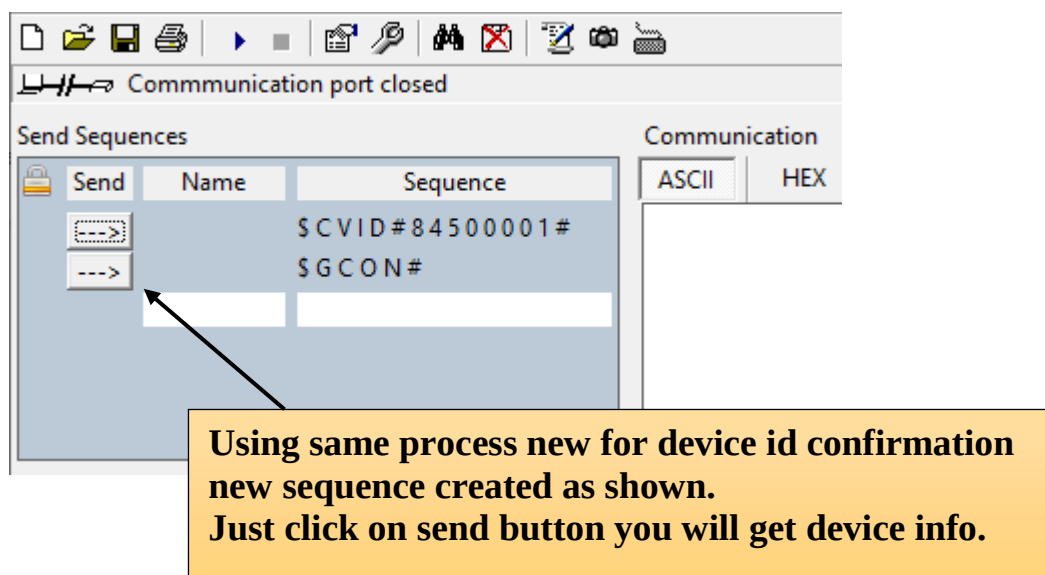


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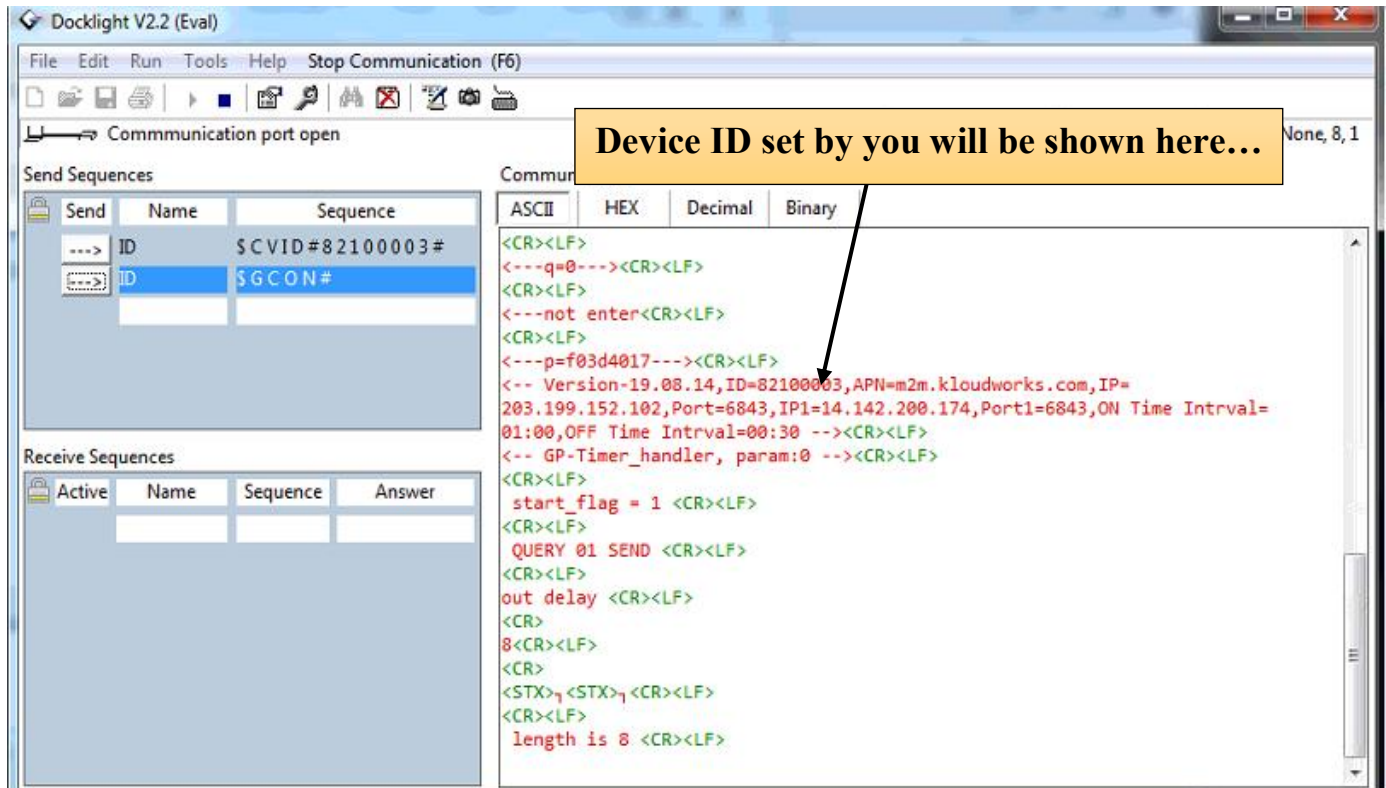
Same as per device ID create sequence for **\$GCON#** this code so you can confirm where device ID set or not.



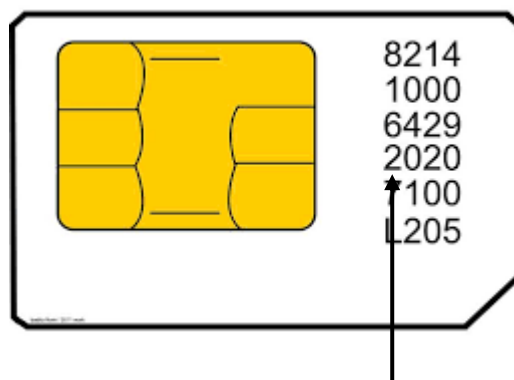
See one more example below...

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Now your device configuration completed just take sim details and site details.



Not down this number with device ID for details.

Now go for installation...